



FAIR 3D HERITAGE

*Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web*

Towards Automating FAIR Compliance in 3D Heritage Models: An AI-Enhanced Epistemological-Operational Framework in the Historic Center of Najaf

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Abstract

The field of digital heritage is undergoing a methodological shift from focusing on the production of 3D models to prioritizing their cognitive structure, interoperability, and long-term sustainability. Despite the expansion of 3D scanning and Heritage Building Information Modelling (HBIM) technologies, many models still suffer from weak metadata, a lack of systematic documentation of modelling decisions, and limited correlation with related data and semantic criteria. This limits their alignment with FAIR principles of discoverability, accessibility, interoperability, and reusability (Wilkinson et al., 2016), as well as with the trust and sustainability requirements of digital repositories (Lin et al., 2020).

This study aims to develop an epistemological-operational framework that utilizes artificial intelligence as an intermediary cognitive structure for reorganizing knowledge within digital heritage models, rather than treating it as a separate analytical tool. The framework was tested in a heritage building within the historic center of Najaf, using architectural data extracted from digital surveys and documentary photographs, and an HBIM model stored in IFC format and interoperable with open data environments. This model is based on the logic of heritage building information modelling as developed in the HBIM literature (Murphy et al., 2013).

The proposed framework consists of four interconnected layers: first, a knowledge extraction layer using machine learning models to classify architectural elements and detect formal patterns; second, a semantic tagging layer based on CIDOC-CRM concepts

and related data to enhance semantic interoperability (Doerr, 2003); third, a paradata organization layer to document modelling decisions in a traceable manner; and fourth, an interoperability testing layer with various digital repositories and presentation platforms. Performance was evaluated by comparing FAIR indices before and after integrating the AI layer, based on metadata completeness, discoverability, number of semantic links, and reusability level.

Preliminary results showed a marked improvement in knowledge organization, data discoverability, and interoperability, as well as enhanced scientific transparency through the systematic documentation of paradata. The study also demonstrated that artificial intelligence (AI) can contribute to addressing some of the institutional challenges associated with post-conflict environments by automating knowledge processes that typically require significant manual effort.

The study's originality lies in redefining AI as an intermediary knowledge architecture within the digital heritage ecosystem, bridging the gap between HBIM, semantic knowledge, and the requirements of digital governance. Through a case study from the Global South within the context of European discussions on digital heritage infrastructure, the study presents a generalizable framework that connects HBIM, AI, and FAIR principles, supporting the development of more sustainable, shareable, and interoperable digital heritage ecosystems.

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